# C PROGRAMMING DEBUGGING TOOLS

# Project Report

Submitted by:- Ranveer Jat

Sap ID :-590032569

Batch:-16

# Tutorial: C Programming - Debugging Tools

## 1. Fundamentals of Errors and Debugging

| Term | Meaning | Example |
|---|---|---|
| **Error** | A mistake in a program that causes it to fail or produce an incorrect result. | Missing ; in C |
| **Bug** | A flaw or mistake in the program's logic that makes it behave incorrectly. | Using + instead of * in a calculation |
| **Debugging** | The process of finding, understanding, and fixing bugs/errors in a program. | Checking the code to find why the calculation is wrong |

## 2. Classifications of Errors in C

| Type of Error | Description | Primary Cause | Example in C |
|---|---|---|---|
| **Syntax Error** | Occurs when the program does not follow the rules/syntax of the C language. | Missing/incorrect symbols, keywords, brackets, semicolons, etc. | printf("Hello") — missing ; |
| **Compilation Error** | Occurs when the compiler cannot convert the source code into object/machine code. | Syntax errors, undeclared variables, incorrect data types, etc. | int x = "Hello"; or using an undeclared variable. |
| **Linker Error** | Occurs after compilation when the linker cannot connect all required definition, missing library, or incorrect linking. | Missing function | Calling calculate() without defining it |

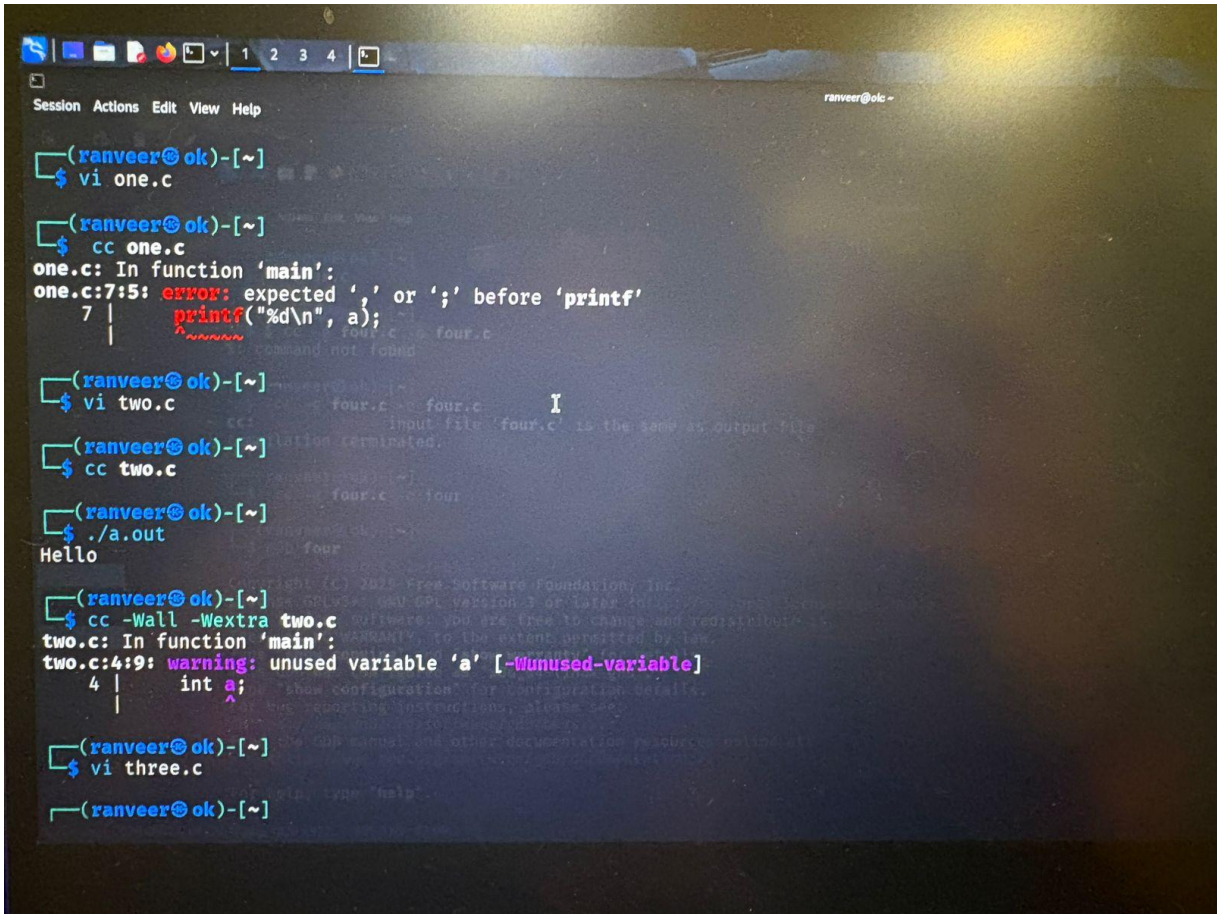
| Type of Error | Description | Primary Cause | Example in C |
| --- | --- | --- | --- |
| | program components. | | |
| **Runtime Error** | Occurs while the program is actually running. | Invalid operations, memory problems, | int x = 10 / 0; division by zero, etc. |
| **Logical Error** | The program runs successfully but produces the wrong result. | Incorrect logic, condition, operation is a * | Using a + b when the formula, required or algorithm. b |

## Q)Why Does a Program Compile Successfully but Produce Incorrect Output?

:- A compiler's job is to verify that your code obeys the syntax and type rules of the C language, and to convert that code into machine instructions. The compiler cannot understand human intent. • If you write sum = a - b; instead of sum = a + b; the statement is completely valid C syntax. Therefore, compilation succeeds, but the resulting logic produces an erroneous output at execution time.

## Case 1: Syntax & Compilation Errors

When a compiler detects a problem, its diagnostic commonly contains the source filename, line number, column number, message category, and explanation. Learning to read these messages is an important programming skill.




**Filename:** one.c

**Line and column:** line 5 & column 5

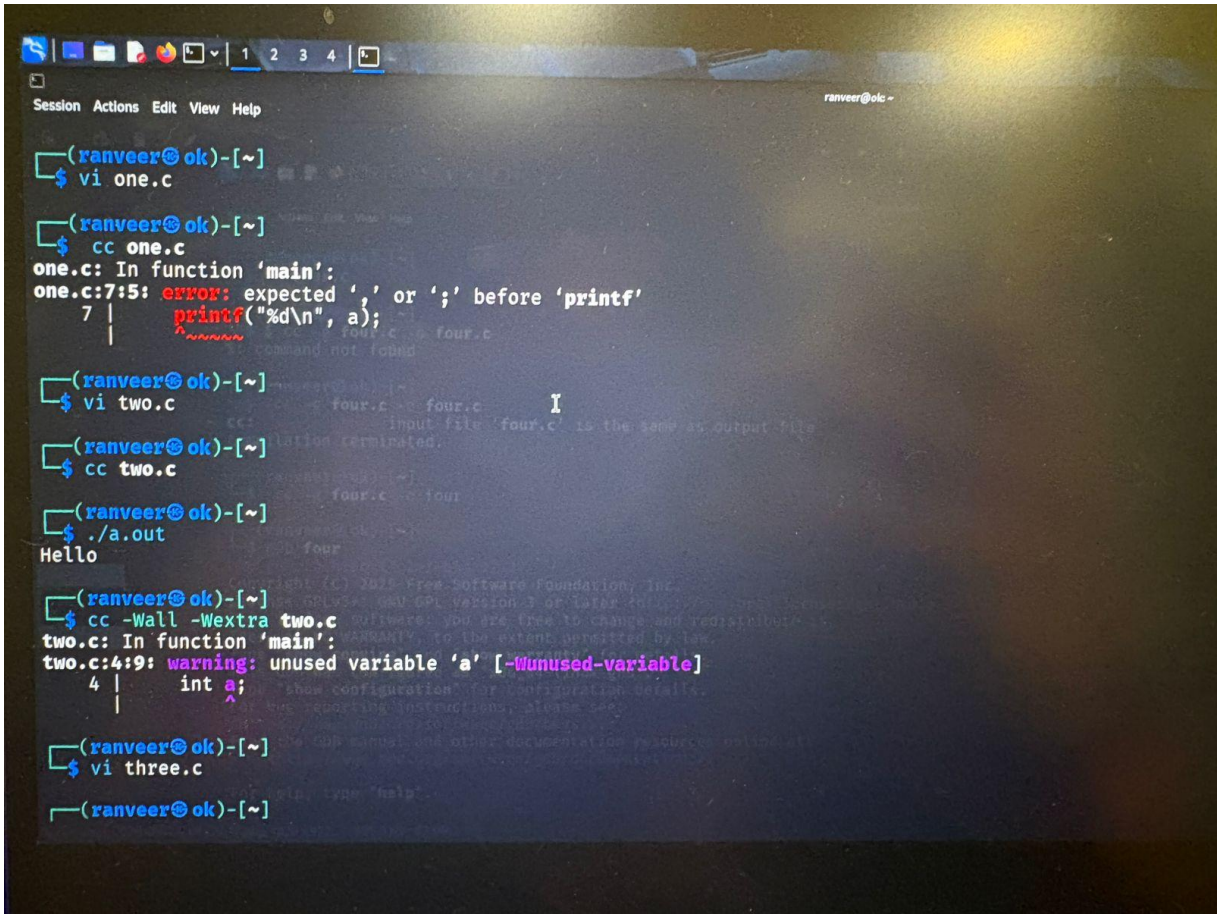
**Error/warning:** indicates the diagnostic severity.

**Description**: Expected , or ; before printf statement.

## Case 2: Compiler Warnings (-Wall, -Wextra)

Warnings are messages about code that the compiler considers suspicious, questionable, or potentially unsafe even though it may still be legal to compile. Warnings are extremely useful because they can expose problems before the program reaches runtime.

**-Wall** enables a broad set of commonly useful warnings. **-Wextra** enables additional warnings that are not all covered by -Wall. For student work and professional C development, compiling with warning options is a valuable habit.




The variable unused may trigger an unused-variable warning when suitable warning flags are enabled. The program may still run, but the warning tells the programmer that the declaration deserves attention.

**Filename:** two.c

**Line and column:** line 3 & column 3

**Error/warning:** indicates the diagnostic severity.

**Description:** gives a clue about what the compiler expected or detected.

## Case-3: printf-Based Debugging (Logical bug)

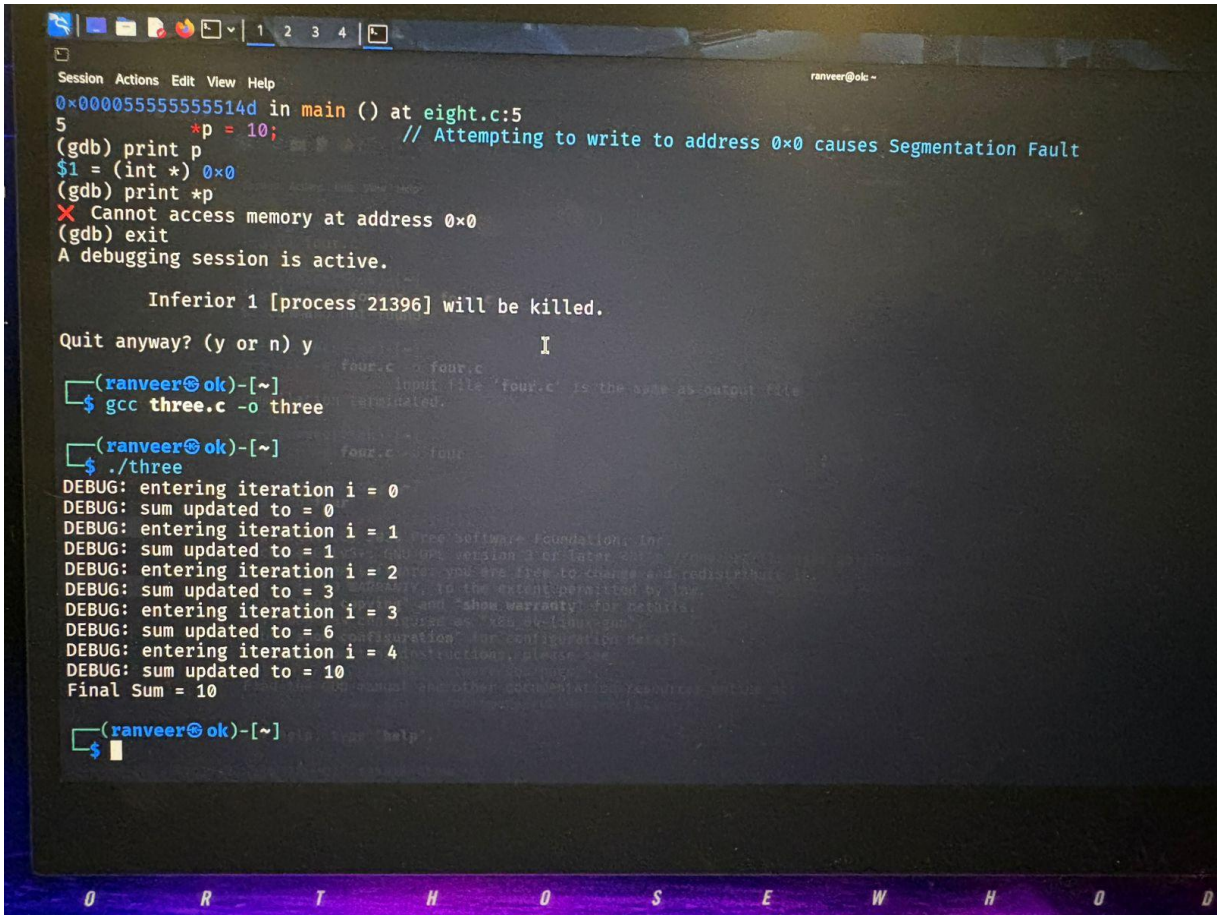
printf-based debugging is a simple method in which temporary output statements are inserted to observe values and execution flow. It is especially useful for beginner programmers because it requires no debugger interface.

**What can be inspected?** Variable values, loop counters, function arguments, intermediate calculations, branches taken, and whether a particular section of code is reached.

printf("DEBUG: a = %d, b = %d\n", a, b); printf("DEBUG: i

= %d, sum = %d\n", i, sum);

For a loop, printing the loop counter and accumulated result can reveal an incorrect starting value, stopping condition, or update expression.




**Limitation:** Too many print statements can make output difficult to read and can sometimes affect timing or program behaviour. Once a program becomes complex, a debugger such as GDB is generally more suitable.

- **Introduction to GDB (GNU Debugger)**
- **GNU:** GNU stands for “GNU's Not Unix!” and is associated with a large free-software project begun by Richard

- Stallman in 1983. GNU provides many tools used in Unix-like development environments.
- **GDB:** The GNU Debugger is a tool for examining a program while it runs. It can pause execution, inspect variables,
- move through statements, examine the call stack, and investigate crashes.
- GDB is particularly useful when a program is too complicated to understand using only print statements. Instead of
- modifying the source repeatedly, the programmer can pause at a selected point and inspect the existing program
- state.

**Installation & Compilation for GDB #**

Update package list and install GDB on Linux systems $ sudo apt update $ sudo apt install gdb # Compile source code with debugging symbols enabled (-g) $ cc -g -Wall -Wextra program.c -o program # Launch GDB with the compiled binary $ gdb ./program.

**Why compile with -g?**

**:-**Without debugging information, GDB can still inspect some lowlevel information, but
source-level debugging becomes much less convenient. With -g,
GDB can associate execution with source lines and make variables easier to inspect.

**GDB Breakpoints**

A breakpoint is a deliberate stopping point in a program. When execution reaches a breakpoint, GDB pauses the
program before executing the selected statement. The programmer can then inspect variables and decide what to do next.
gdb ./program (gdb)
break main
(gdb) run
Breakpoints can be placed at function names or source lines. They are useful when you know approximately where the incorrect behavior occurs and want to inspect the program immediately before it happens.

**Example:** If a variable becomes incorrect inside a loop, place a breakpoint at the statement that updates it. Inspect
the variable, continue execution, and inspect it again on later iterations.

**Important GDB Commands – Detailed Guide**

**run (r):** Starts program execution from the beginning or restarts it. **break (b):** Creates a breakpoint. **print (p):** Displays the current value of an expression. **display:** Automatically displays an expression after stops/steps. **continue (c):** Resumes execution until another stopping point.

**list (l):** Shows source code around the current location. **info locals:** displays local variables in the current stack frame.

**backtrace (bt):** displays the call stack. **info breakpoints:** shows breakpoints and their status. **delete:** removes a breakpoint. **disable/enable:** temporarily turns a breakpoint off or on.

**quit:** exits GDB.

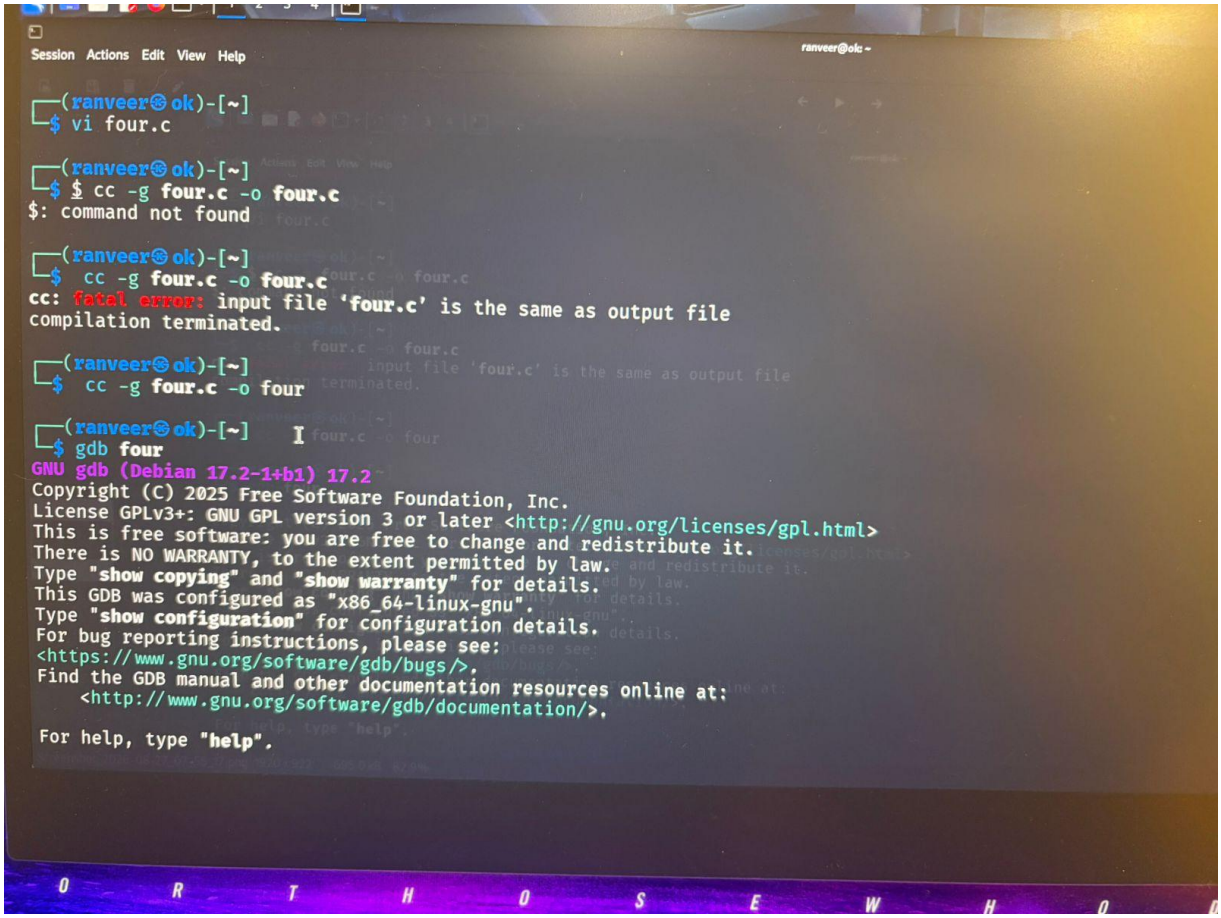
**Why commands matter:** A debugger is most effective when you use commands with a clear question in mind.

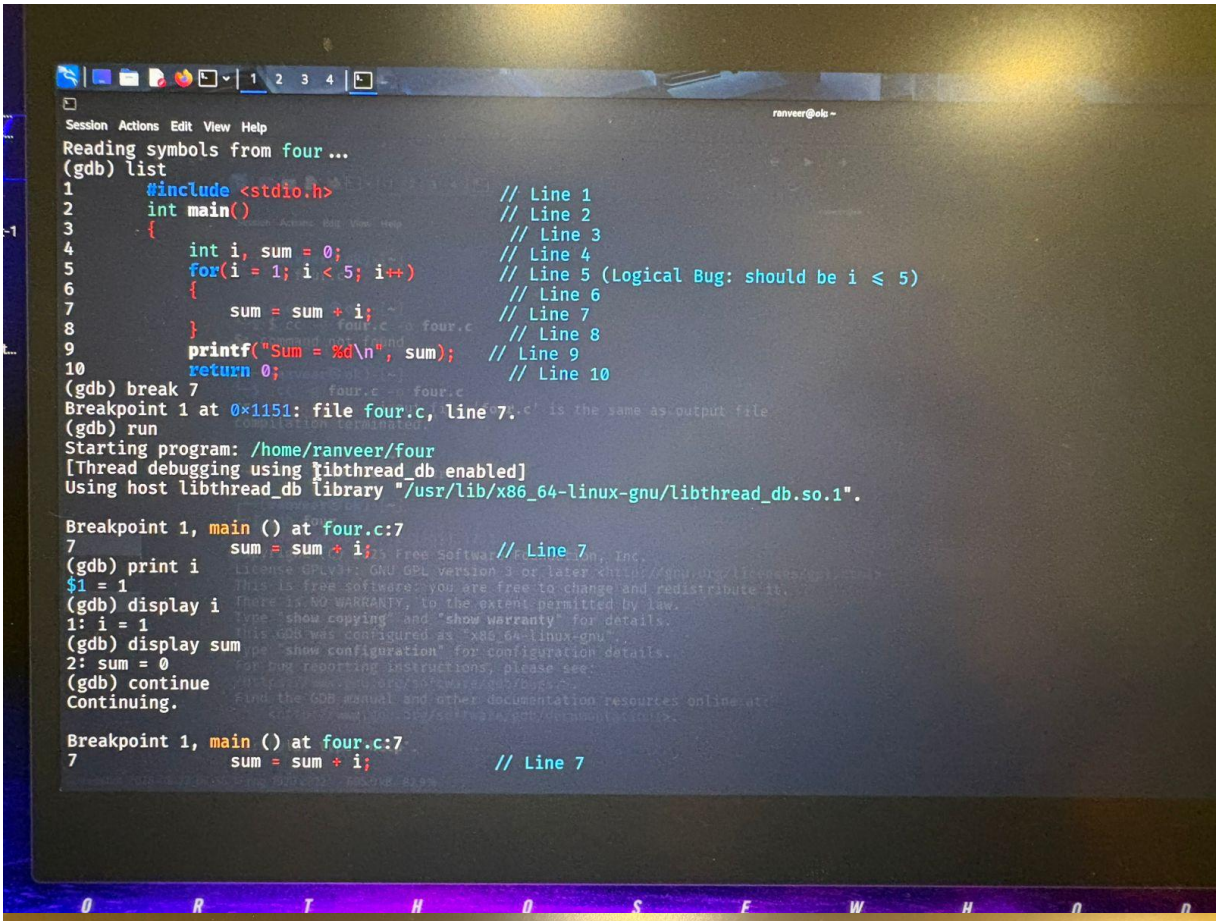
For example, use print i to answer “What is the loop counter right now?” rather than stepping randomly through the program.

## Case 4: Debugging a Logical Error using GDB

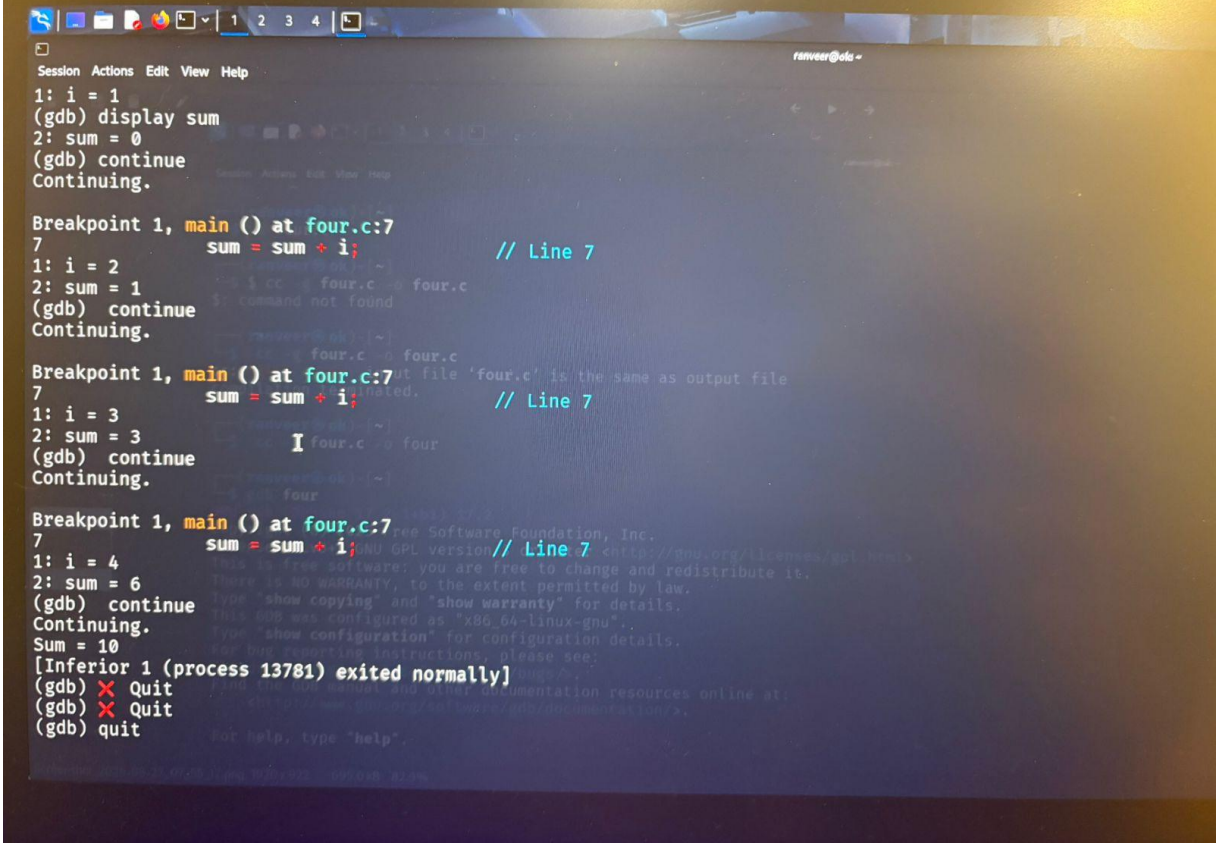
- Expected Output: 15 (1+2+3+4+5)

- Actual Output: 10 (1+2+3+4)



- 

```
Session Actions Edit View Help
Reading symbols from four...
(gdb) list
1       #include <stdio.h>                    // Line 1
2       int main()                           // Line 2
3       {                                     // Line 3
4           int i, sum = 0;                  // Line 4
5           for(i = 1; i < 5; i++)           // Line 5 (Logical Bug: should be i <= 5)
6           {                                 // Line 6
7               sum = sum + i;               // Line 7
8           }                                 // Line 8
9           printf("Sum = %d\n", sum);       // Line 9
10          return 0;                         // Line 10
(gdb) break 7
Breakpoint 1 at 0x1151: file four.c, line 7.
(gdb) run
Starting program: /home/ranveer/four
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at four.c:7
7               sum = sum + i;               // Line 7
(gdb) print i
$1 = 1
(gdb) display i
1: i = 1
(gdb) display sum
2: sum = 0
(gdb) continue
Continuing.

Breakpoint 1, main () at four.c:7
7               sum = sum + i;               // Line 7
```

- 

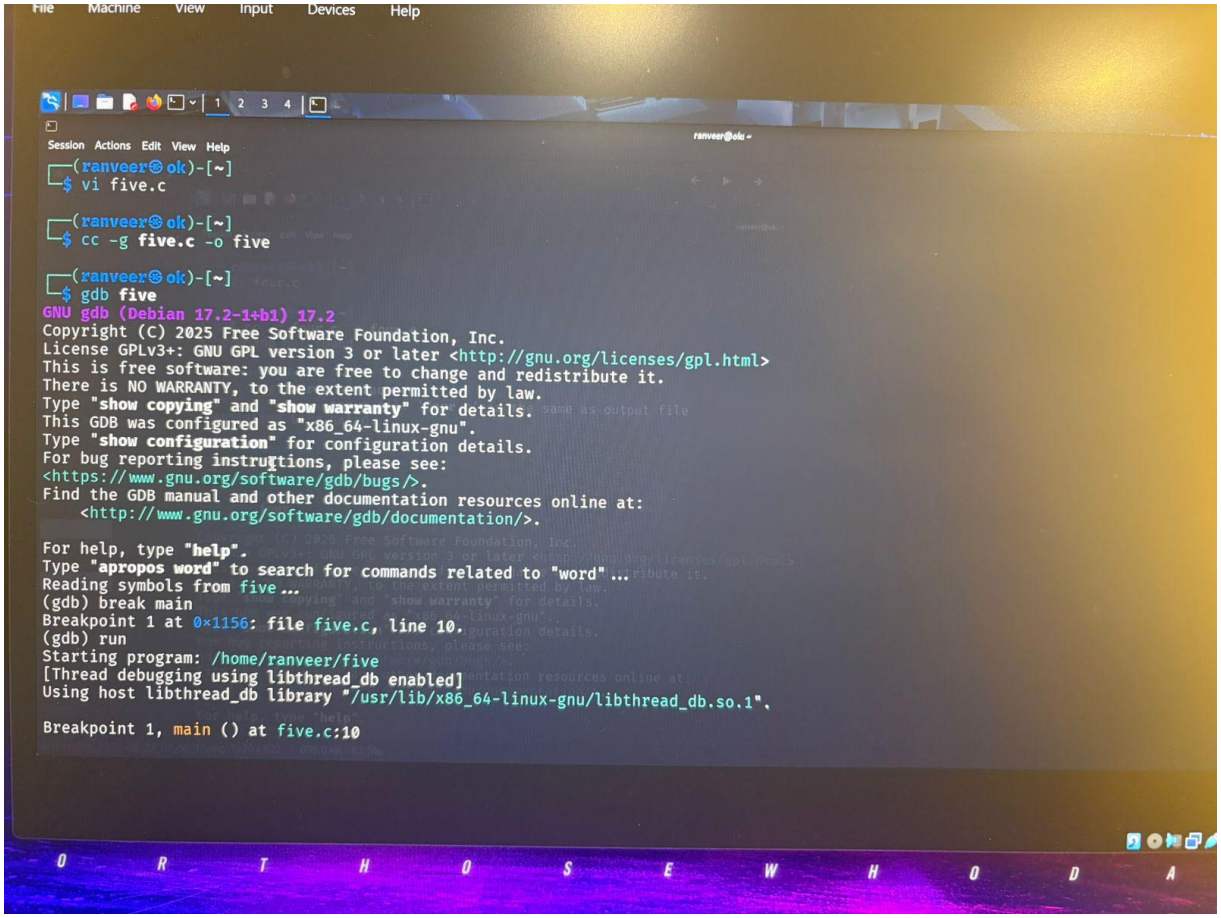
```
Session Actions Edit View Help
1: i = 1
(gdb) display sum
2: sum = 0
(gdb) continue
Continuing.

Breakpoint 1, main () at four.c:7
7               sum = sum + i;               // Line 7
1: i = 2
2: sum = 1
(gdb)  continue
Continuing.

Breakpoint 1, main () at four.c:7
7               sum = sum + i;               // Line 7
1: i = 3
2: sum = 3
(gdb)  continue
Continuing.

Breakpoint 1, main () at four.c:7
7               sum = sum + i;               // Line 7
1: i = 4
2: sum = 6
(gdb)  continue
Continuing.
Sum = 10
[Inferior 1 (process 13781) exited normally]
(gdb) Quit
(gdb) Quit
(gdb) quit
```

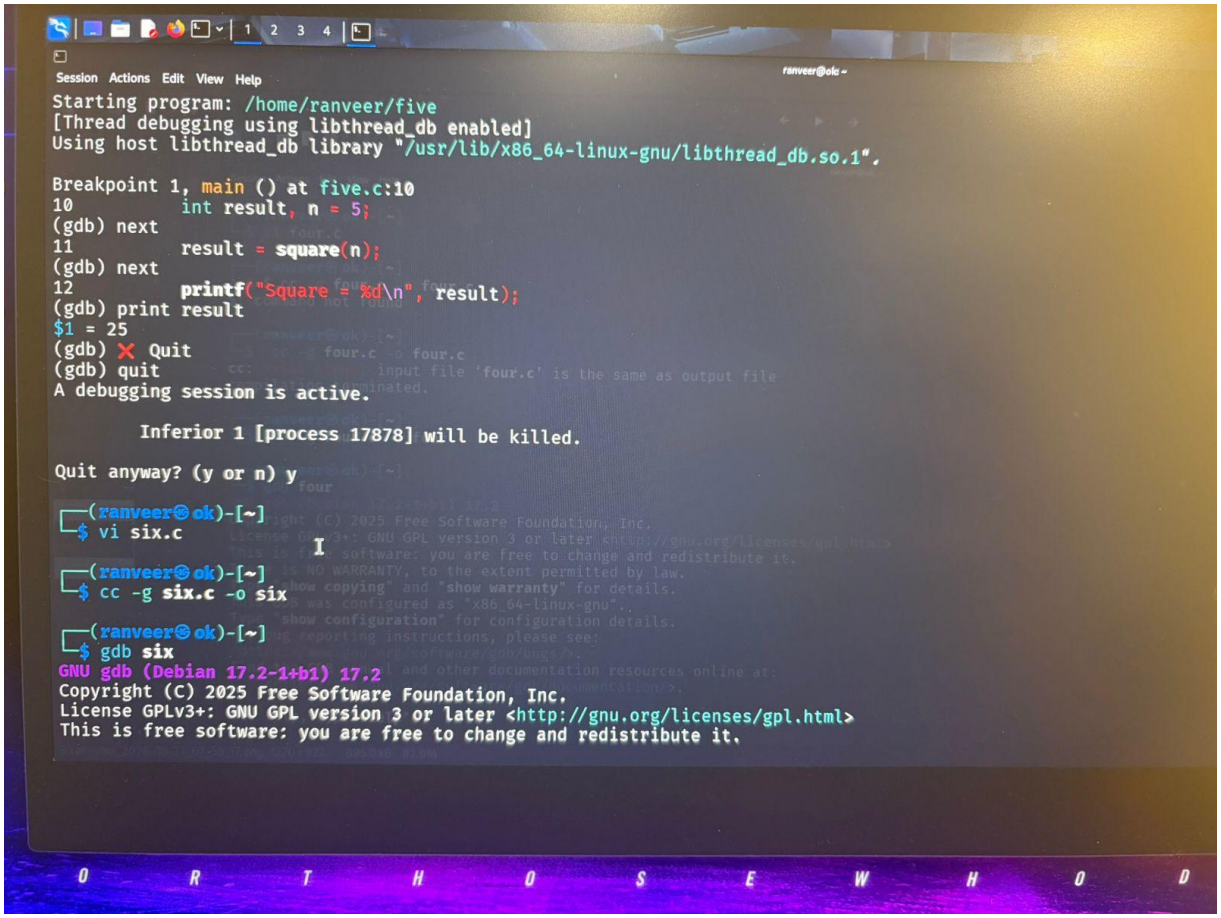
File name-four.c

## Case 5: Debugging Functions (next vs. step)

Both commands move execution forward, but they differ when the current line contains a function call.

**next (n):** steps over a function call. If the current line calls a function, GDB executes the function and returns to the caller without entering the function body line-by-line.



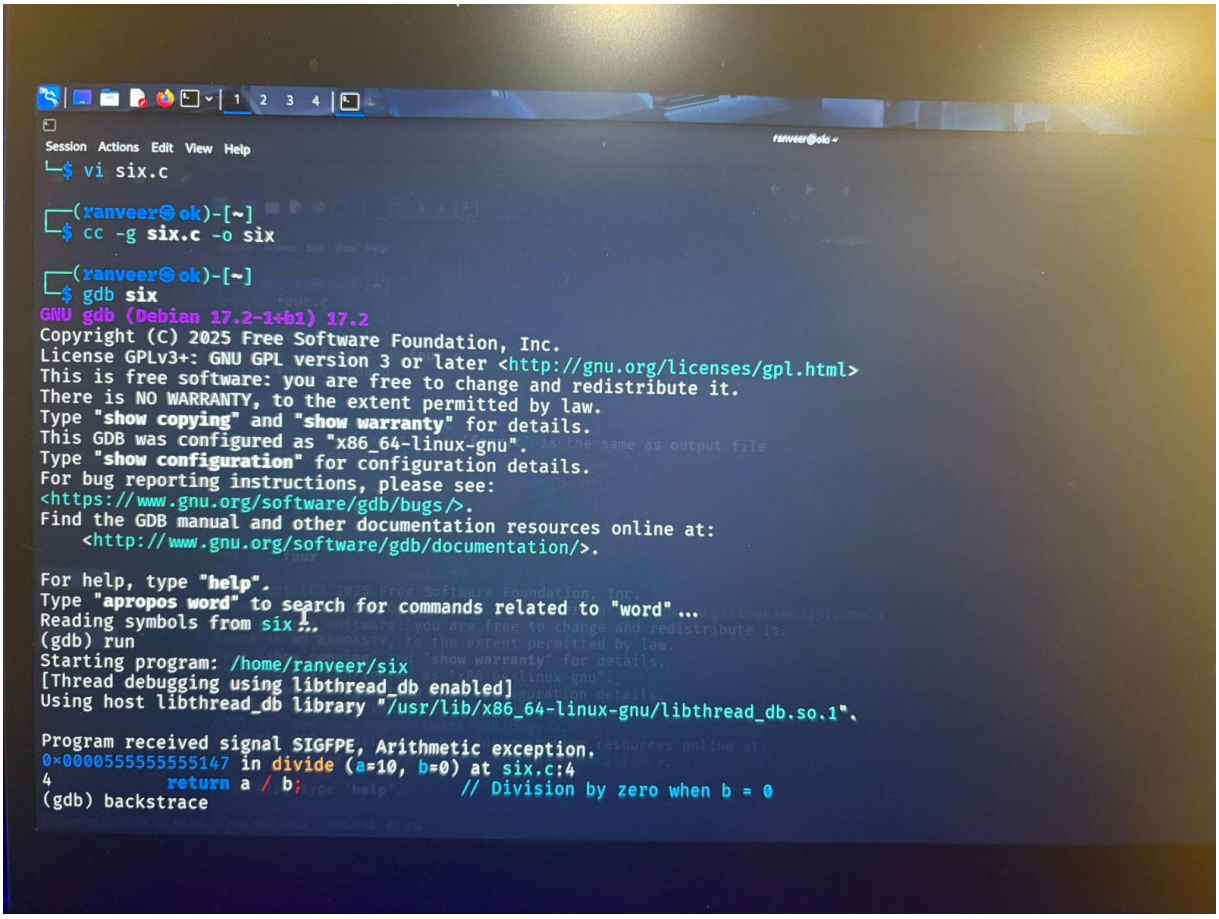


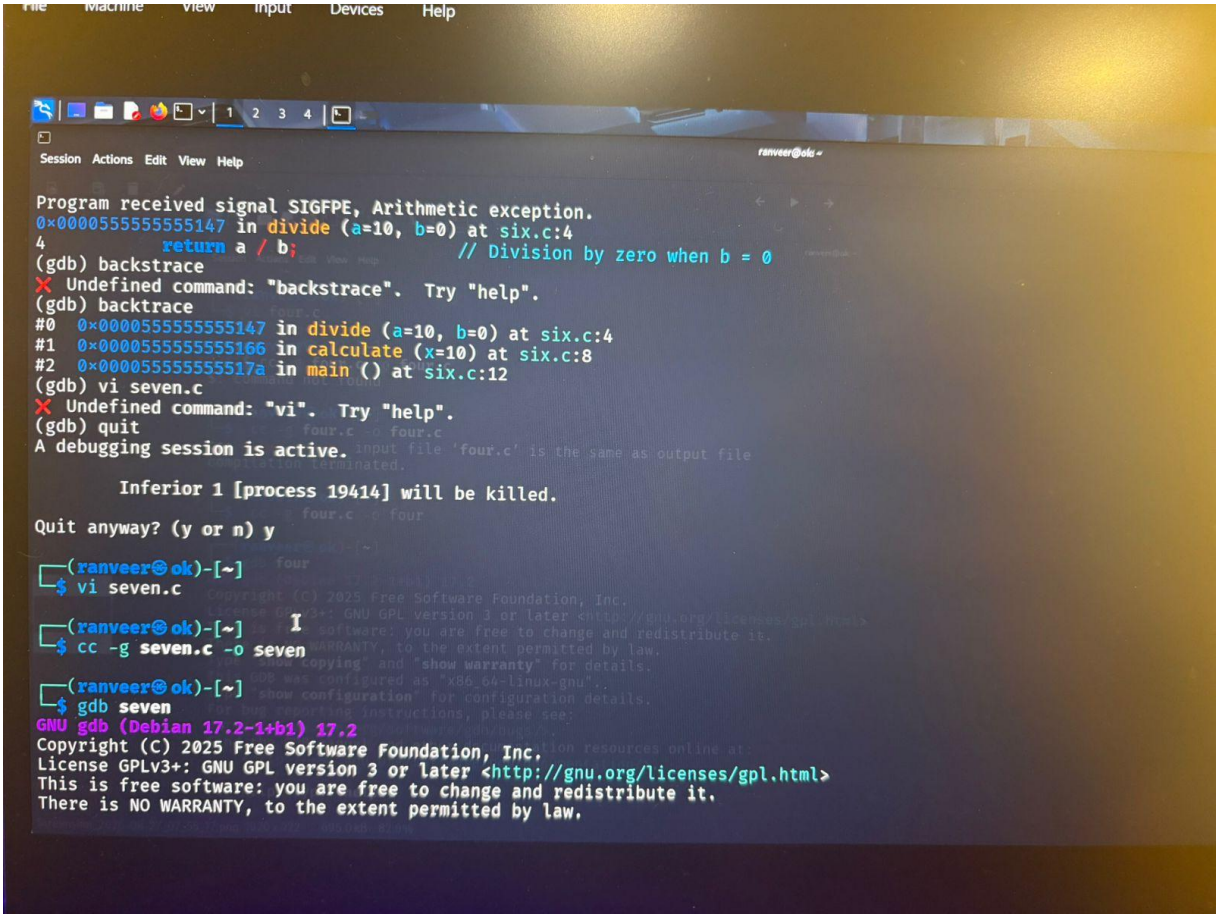

**step (s):** steps into a function call. GDB enters the called function, allowing you to inspect its parameters, local variables, and statements.

If you trust square() and only want its result, use next. If you suspect the multiplication or the function's internal state, use step.

## Case 6: Debugging Runtime Errors using Backtrace (backtrace / bt)

When a program crashes (e.g., floating point exception / division by zero), GDB can pinpoint the exact function call sequence leading to the crash.





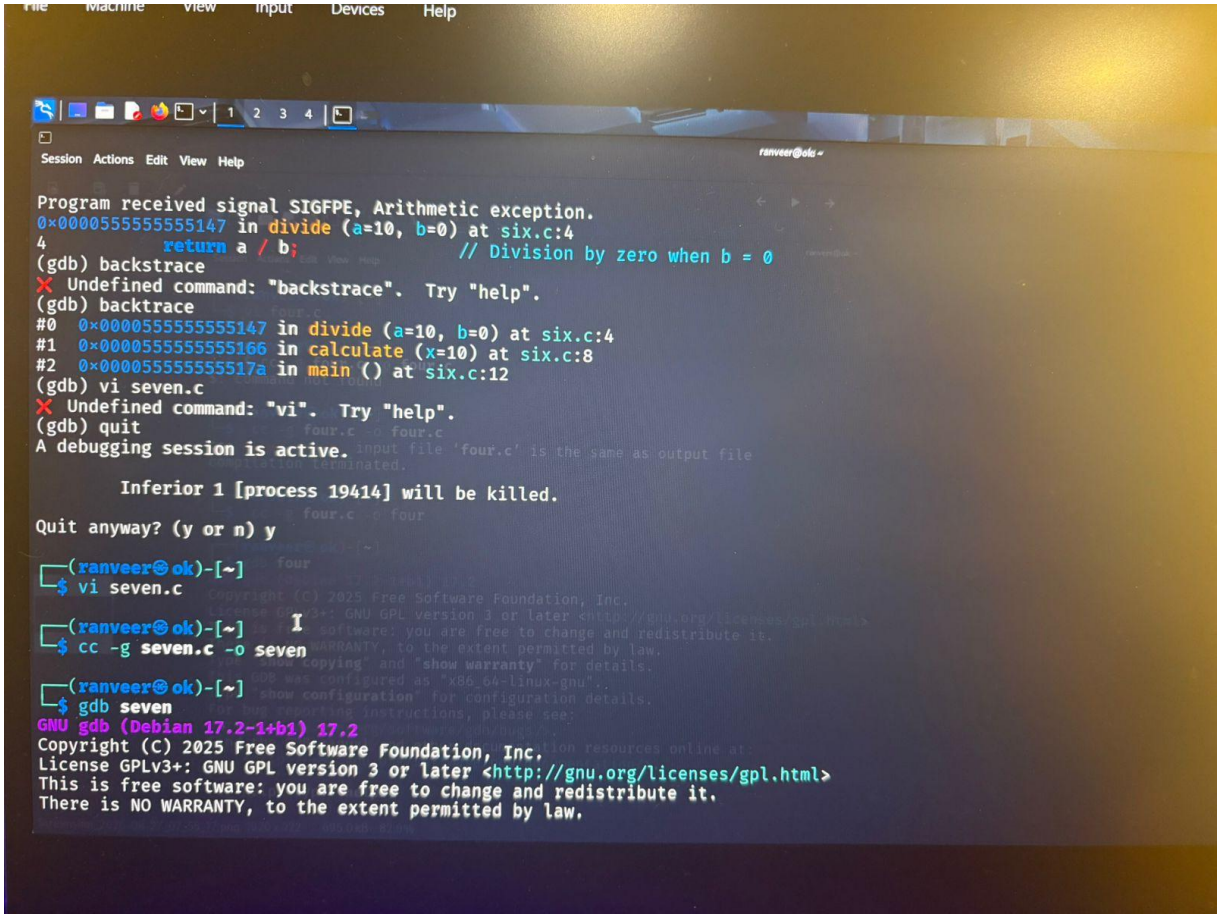

**Understanding Stack Frames**

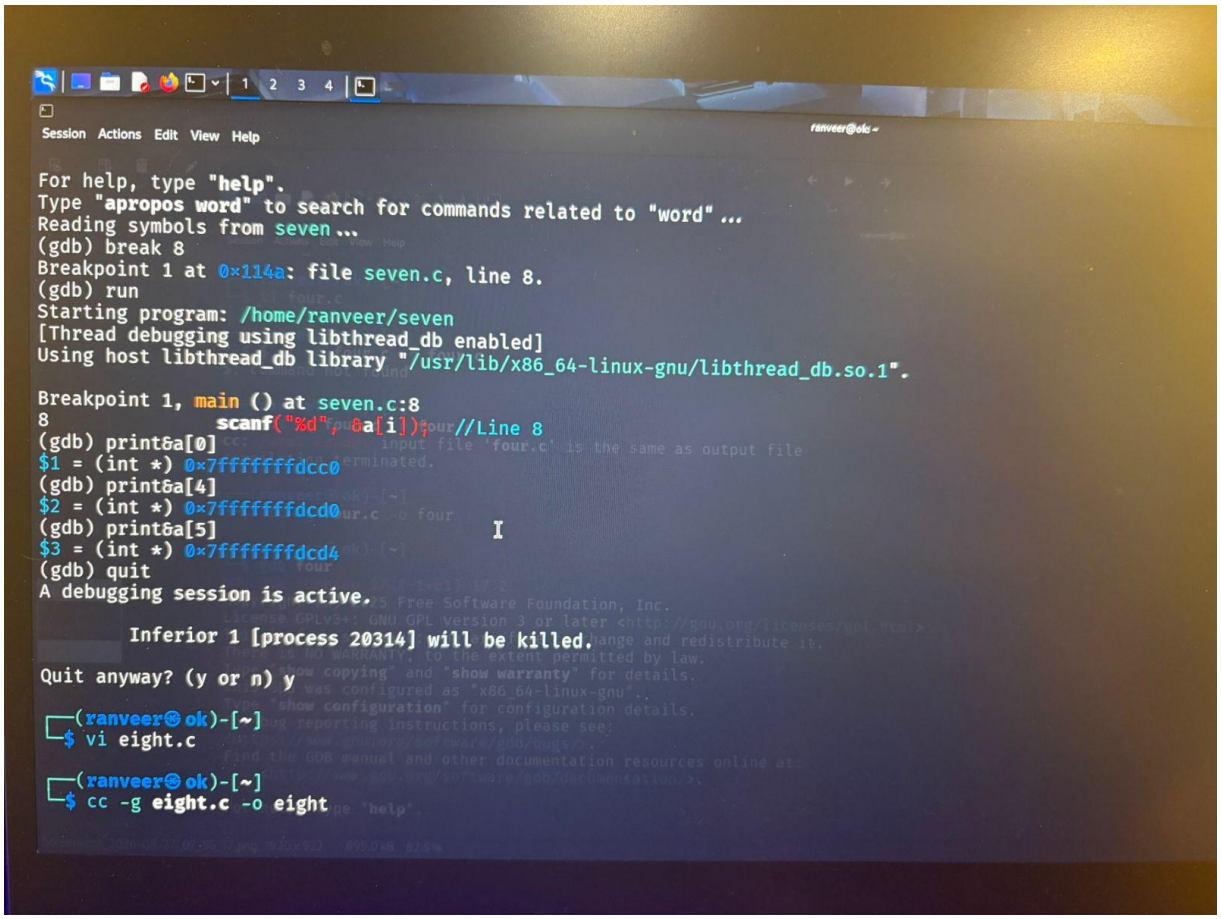
- Frame #0: The function where the crash actually occurred (divide).
- Frame #1: The function that called frame #0 (calculate).
- Frame #2: The top-level function that started the call sequence (main).

## Case 7: Debugging Array Out-of-Bounds Errors using GDB

C arrays have fixed storage determined by their declared size, but C does not automatically perform bounds checking for ordinary array indexing. For int a[5], only indices 0 through 4 belong to the array. int a[5];

The loop executes for i = 0, 1, 2, 3, 4, and 5. The access a[5] is outside the declared array. This is undefined behaviour and may overwrite unrelated memory or cause other failures.


File Machine View Input Devices Help
Session Actions Edit View Help
Program received signal SIGFPE, Arithmetic exception.
0x0000555555555147 in divide (a=10, b=0) at six.c:4
4 return a / b; // Division by zero when b = 0
(gdb) backstrace
Undefined command: "backstrace". Try "help".
(gdb) backtrace
#0 0x0000555555555147 in divide (a=10, b=0) at six.c:4
#1 0x0000555555555166 in calculate (x=10) at six.c:8
#2 0x000055555555517a in main () at six.c:12
(gdb) vi seven.c
Undefined command: "vi". Try "help".
(gdb) quit
A debugging session is active.
Inferior 1 [process 19414] will be killed.
Quit anyway? (y or n) y
(ranveer㉿ok)-[~]
$ vi seven.c
(ranveer㉿ok)-[~]
$ cc -g seven.c -o seven
(ranveer㉿ok)-[~]
$ gdb seven
GNU gdb (Debian 17.2-1+b1) 17.2
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.


Session Actions Edit View Help
For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from seven...
(gdb) break 8
Breakpoint 1 at 0x114a: file seven.c, line 8.
(gdb) run
Starting program: /home/ranveer/seven
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/x86_64-linux-gnu/libthread_db.so.1".
Breakpoint 1, main () at seven.c:8
8 scanf("%d", &a[i]); //Line 8
(gdb) print&a[0]
$1 = (int *) 0x7fffffffdcc0
(gdb) print&a[4]
$2 = (int *) 0x7fffffffdcd0
(gdb) print&a[5]
$3 = (int *) 0x7fffffffdcd4
(gdb) quit
A debugging session is active.
Inferior 1 [process 20314] will be killed.
Quit anyway? (y or n) y
(ranveer㉿ok)-[~]
$ vi eight.c
(ranveer㉿ok)-[~]
$ cc -g eight.c -o eight

**Correct loop:**

for (i = 0; i < 5; i++) { scanf("%d",

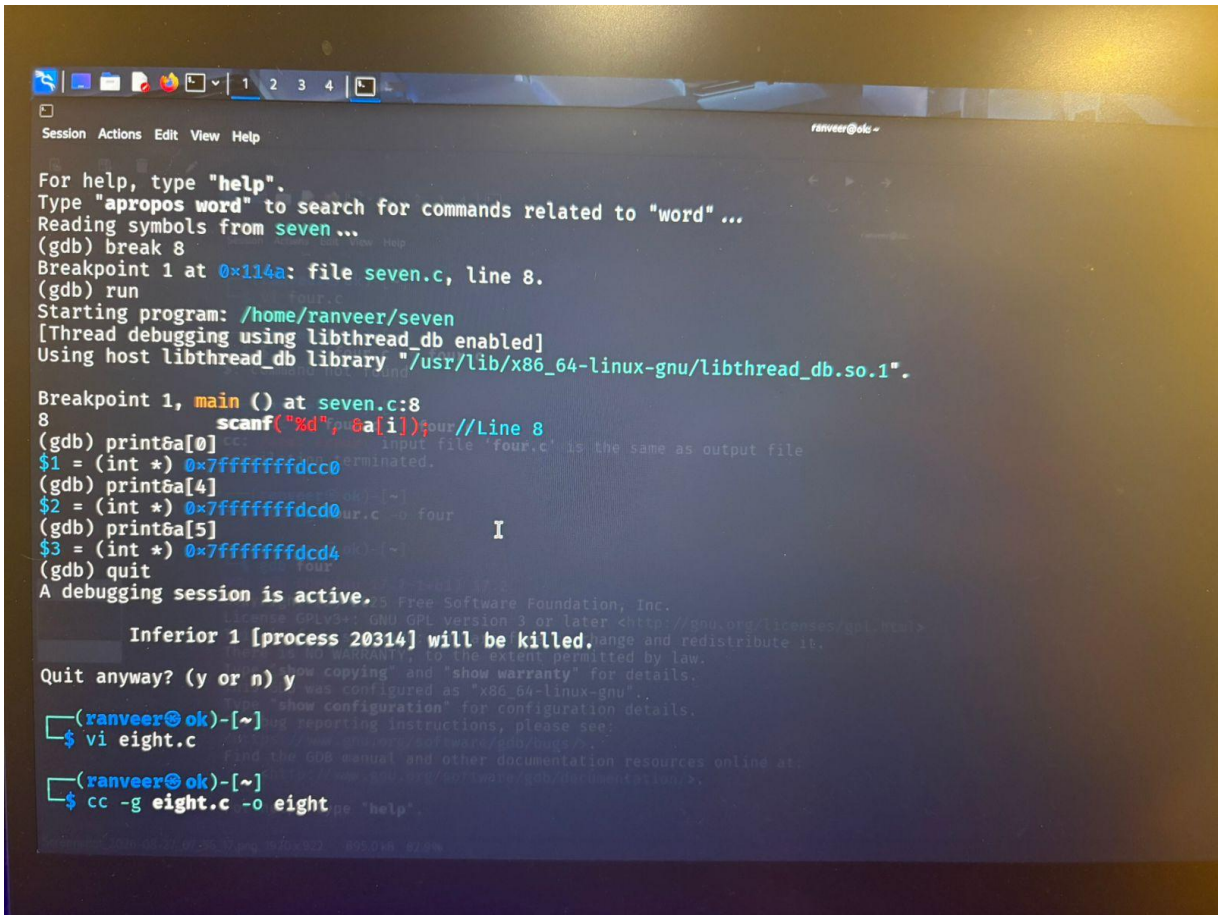
&a[i]);

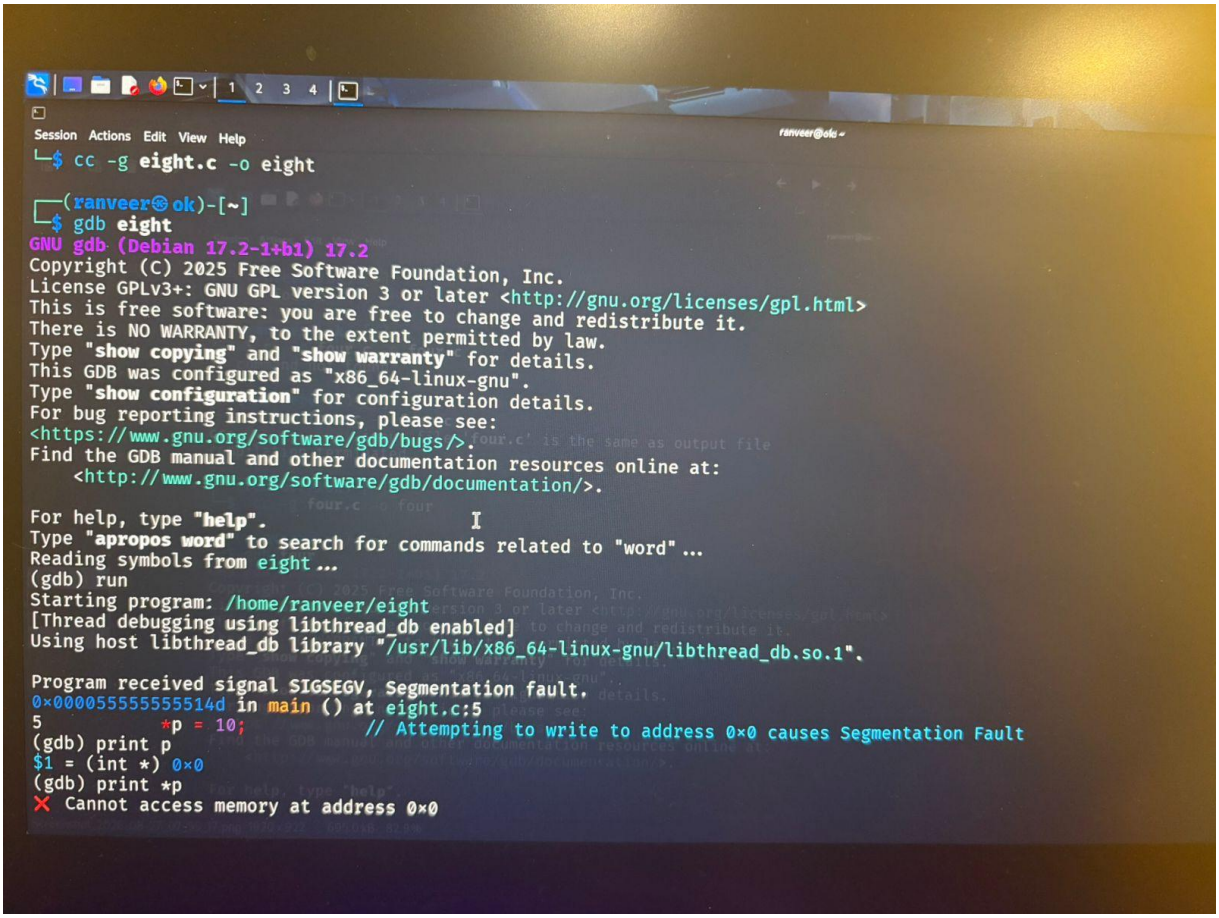
}

**Debugging approach:** inspect the loop condition, print the index, place a breakpoint at the array access, and check whether the index is within the valid range before the access occurs.

## Case 8: Segmentation Faults (SIGSEGV) using GDB

A segmentation fault is a common operating-system-level indication that a process attempted an invalid memory access. One common cause is dereferencing a NULL pointer.






The pointer p contains a null address. Dereferencing it with *p attempts to access memory through that invalid pointer.

cc -g sample8.c -o sample8 gdb

./sample8

(gdb) run

Program received signal SIGSEGV, Segmentation fault.

(gdb) print p

(gdb) print *p

GDB may show p = 0x0. Attempting to print *p can itself fail because the pointed-to memory is not valid.

**Corrected approach:**

```
int x; int *p = &x;

*p = 10;

printf("%d\n", *p);
```

Now p points to the valid storage occupied by x, so writing through p is valid.

## Systematic Eight-Step Debugging Strategy

1. **Observe expected vs actual output** — Write down what the program should produce and what it actually produces. Avoid vague statements such as "it is not working."

2. **Reproduce the issue consistently** — Find inputs and conditions that reliably trigger the bug. Consistent reproduction makes investigation much easier.

3. **Compile with -Wall -Wextra -g** — Warnings may reveal suspicious code, while -g provides information for source-level GDB debugging.

4. **Isolate suspicious code** — Reduce the problem to the smallest relevant function, loop, condition, or statement.

5. **Inspect state** — Use printf for simple tracing or GDB commands such as break, print, display, and info locals.

6. **Apply one minimal fix** — Change one relevant thing at a time. Large uncontrolled changes make it difficult to know what actually solved the problem.

7. **Recompile and re-test** — A source change must be rebuilt. Then run the same test that originally exposed the problem.

8. **Verify edge cases and boundaries** — Test zero, negative values, maximum/minimum values, empty input,
first/last loop iterations, and other boundary conditions as appropriate.

## Final Revision Summary

The most important idea is that different errors appear at different stages. Syntax and many type/declaration problems are found during compilation. Linker errors occur when compiled components cannot be combined.

Runtime errors appear while the executable is running. Logical errors are different because the program may run normally while still doing the wrong thing.

Debugging is therefore an evidence-based process. Start with the expected and actual behavior, reproduce the problem, inspect warnings, isolate the suspicious code, inspect variables and execution flow, make a controlled correction, rebuild, and test again. For simple problems, printf tracing can be enough; for deeper problems, GDB provides breakpoints, stepping, variable inspection, stack-frame analysis, and crash investigation.

**One-line memory trick:** *Compiler finds many code mistakes ® Linker connects program parts ® Runtime*

*reveals execution failures ® Logic testing reveals wrong answers ® Debugging finds and fixes the underlying problem.*